

Wearable Robotics (AME 60553)

Lecture 1: Introduction, nomenclature,
and motivation

Welcome!

Prof. Edgar Bolívar

Aug. 23rd, 2023

The big picture

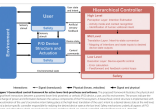
How do we measure joint angles, joint moments, and muscle activity from experimental data?

Biomechanics and muscular activity during sit-to-stand transfer

M.E. Roelbroeck MSc¹, C.A.M. Doorenbosch MSc^{1,2}, J. Harlaar MSc¹, R. Jacobs PhD¹, G.J. Lankhorst MD PhD¹
¹Department of Rehabilitation Medicine, Free University Hospital, and ²Department of Functional Anatomy, Faculty of Human Movement Sciences, Free University, Amsterdam, The Netherlands



Control strategies for active lower extremity prosthetics and orthotics: a review
Tobias et al.
© Elsevier

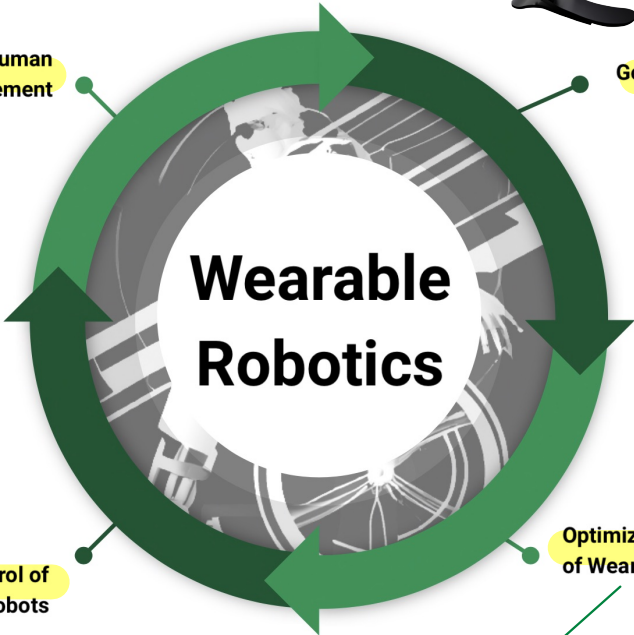


The 3-layer approach
|
Impedance control for low-level control
|
Phase variables for mid-level control

Kinematics
Kinetics
Hill-Type muscle models
Analysis of Human Movement



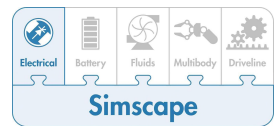
OpenSim



*With high specific torque
high-performance electric brushless motors

Generation of Movement

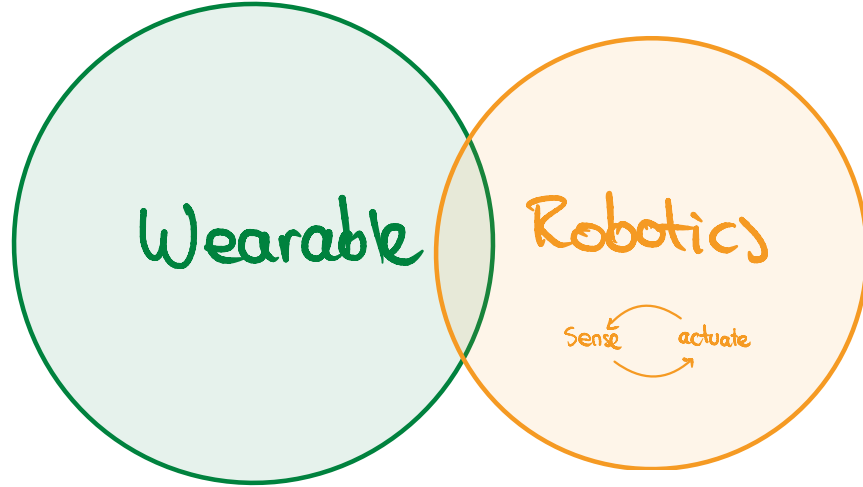
The role of elastic components in series and parallel with the motor



Optimization in the Context of Wearable Robots

Convexity - what, why, how?
Dealing with uncertainty
Real-time optimization

Disclaimer: Always start by understanding the person's goals and challenges!



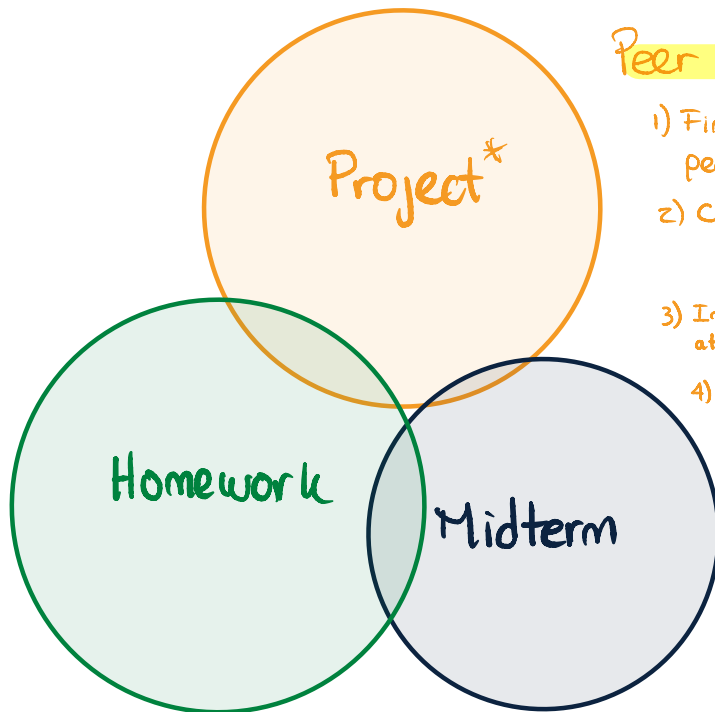
The logistics



Alison Bryson Ph.D.
Research Scientist at
ILMC
Dec. 6th



Ray Browning Ph.D.
CEO & Co-Founder
BIOMESTY, Inc.
Dec. 9th



Challenging
Assignments!

(Partially done
in our Wednesday
sessions)

Peer advisor

- 1) Find a project to peer advise
- 2) Critique*
Strengths and weaknesses of the approach
- 3) Incomplete critiques will affect the peer advisor's grade
- 4) Share the draft of your milestone with your peer advisor ^{* Critique from peer-advisor will attach your draft.} at least one week before the deadline.
- 5) Receive critique from peer advisor at least one day before the deadline.
Discuss the critique in-person for at least 10 minutes before the submission.

*Critique (Merriam-Webster)

noun

critique | \ ˈkraɪ-ték ④, kri-\

Definition (Entry 1 of 2)

: an act of criticizing

especially : a critical (see CRITICAL sense 1c) estimate or discussion

// a critique of the poet's work

// an honest critique of her art

verb

critiqued; critiquing

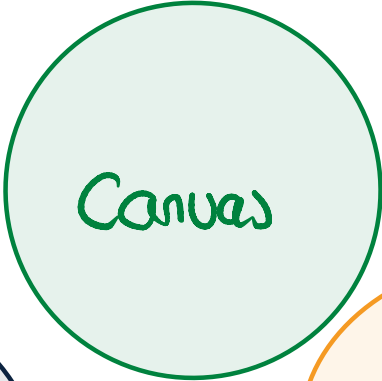
Definition (Entry 2 of 2)

transitive verb

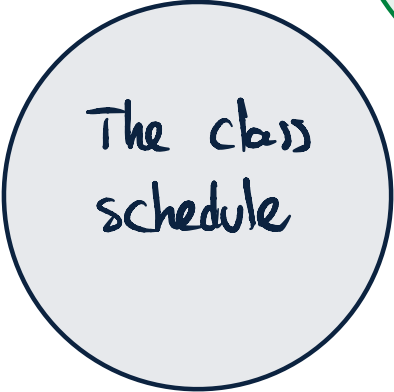
: to examine critically : REVIEW

// critique the plan

Let's check:



Canvas



The class
schedule

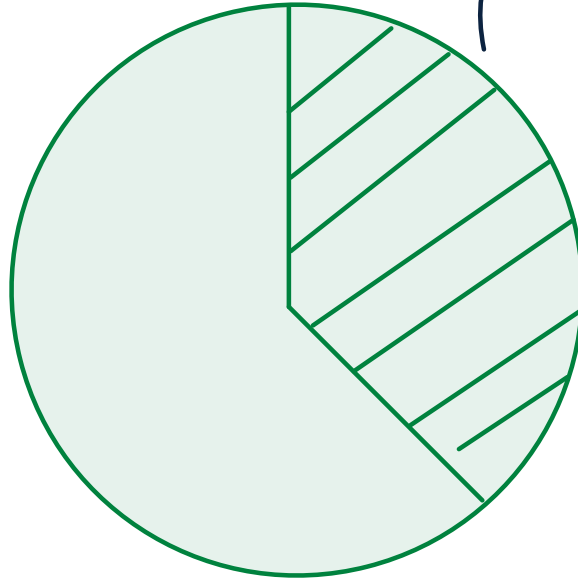


The syllabus

Meet your classmates

5 graduate students

4 AME, 1 BIOC



3 undergraduate students

ME

- Introduction
- Motivation to take the class
- Expectations from the class
- Hobbies

Meet your instructor:

Edgar Bolívar-Nieto, Ph.D.

Principal Investigator

Assistant Professor

Aerospace and Mechanical Engineering

Areas of Interest

Dr. Bolívar-Nieto researches the optimal control, state estimation, and actuator design of wearable robots. He is interested in torque-dense actuators that, combined with elastic elements and intelligible control algorithms, enable wearable technology that is comfortable and intuitive to use and promote clinical outcomes for people with disabilities.

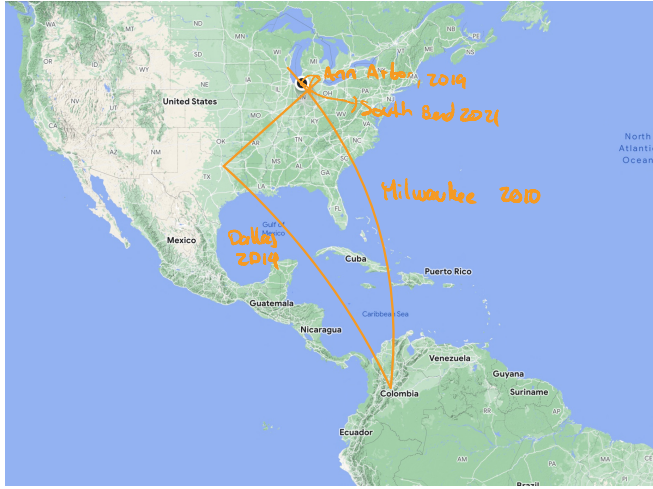
Google Scholar 



Office 368 Fitzpatrick Hall,
Notre Dame, IN
46556 USA

Phone 574-631-9516

Email ebolivar@nd.edu



Meet your TA:

David Kelly

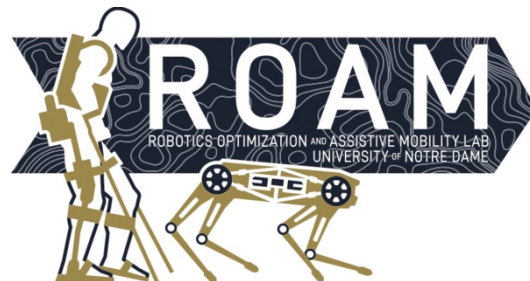
Graduate Student

Education

B.S. University of Notre Dame (2018)

Research

My research interests lie at the intersection of biomechanics, robotics control, and healthcare. I am particularly invested in how the use of robotics and mechanical systems can enhance the lives of individuals with medical needs.



Office B22 Fitzpatrick Hall

Email dkelly7@nd.edu

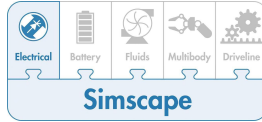
MATLAB:

MATLAB Coder: Generate C or C++ code from your source MATLAB code. Use "codegen"

Simulink:

- Visual representation of text code
- Useful to interconnect systems of multiple sample rates

Simscape:



- "Easier" to interconnect hardware-in-the-loop

OpenSim:



Python:

Community: make your code accessible to everyone

Use the right tools: Visual Studio Code

We will not use Python but questions/comments are highly encouraged

Introduction ✓

Nomenclature

Motivation

Notation: Equations

Differential equation: it is an equation relating derivatives of one or more variables, e.g.,

$$\ddot{x} = x, \quad \text{or} \quad \frac{dx}{dt} = 0 + y$$

* $\dot{x} = \frac{dx}{dt}$ ↗ Notation

Ordinary differential equation (ODE): it is a differential equation where all the derivatives are with respect to a single independent variable, e.g.,

$$\ddot{x} = \sin(x), \quad \frac{dx}{dt} + \frac{dx}{dy} = 3$$

Algebraic equation: statement of the equality of two algebraic expressions

it is a differential equation but it is not ordinary

For example:

$x^3 + 1 = 3$, $x = \sin(x)$
Algebraic $\uparrow$ $\uparrow$ Not algebraic.

addition, subtraction, multiplication, division, powers

Notation: Linear algebra and functions

$\mathbb{R}$: set of real numbers $\rightarrow$ scalars parameters will be normally represented by Greek letters $(\alpha, \beta, \dots)$

$\mathbb{R}^n$: set of n -dimensional vectors with real numbers

$\hookrightarrow$ vectors will be represented by lower-case letters: $a \in \mathbb{R}^n$ means

" a belongs to the set of n -dimensional vectors."

$\mathbb{R}^{n \times m}$: set of $n \times m$ matrices

n : # rows

m : # columns;

$\hookrightarrow$ represented by capital letters (eg, A, B, C)

For example,

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 5 & 6 \end{bmatrix}, A \in \mathbb{R}^{2 \times 3}$$

For example:

$$a = \begin{bmatrix} 1 \\ 5 \end{bmatrix}, a \in \mathbb{R}^2$$

$\mathbf{1}$: Can be the regular 1 or a column vector with 1 in all its elements.

Notation: Functions

* Mathematical notation
thanks to "convex optimization"
book. - Boyd and Vandenberghe

When we write

$$f: A \rightarrow B$$

We mean

f is a function on the set $\text{dom } f \subseteq A$
into the set B

↗ domain of f

For example:

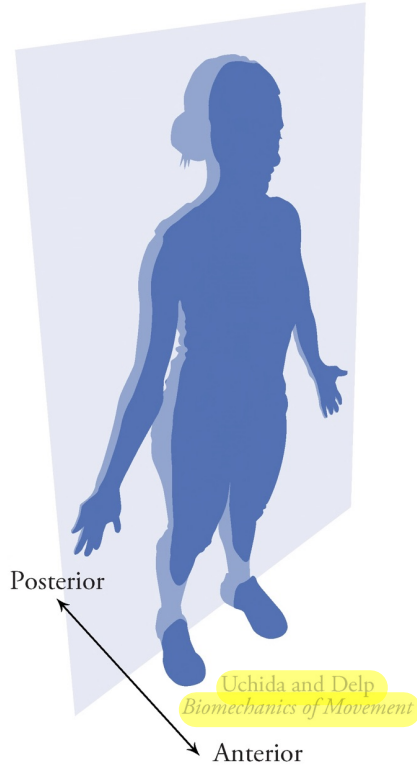
$$f(x) = \begin{bmatrix} \frac{1}{x} \\ x \end{bmatrix}; \quad f: \mathbb{R} \rightarrow \mathbb{R}^2$$

→ Similar to specifying
data types in programming

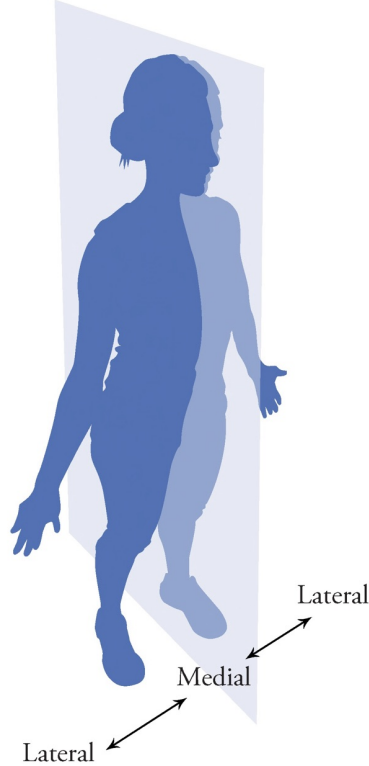
* Note that $\text{dom } f$ is every real number except
0 because $1/x$ is not defined then.

Notation: Anatomical planes and directions

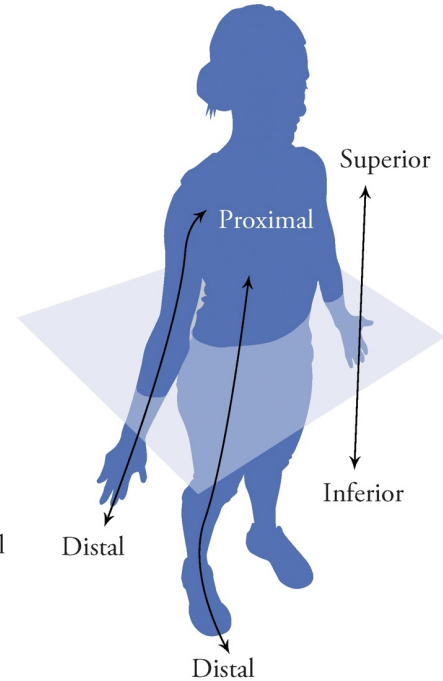
Frontal plane

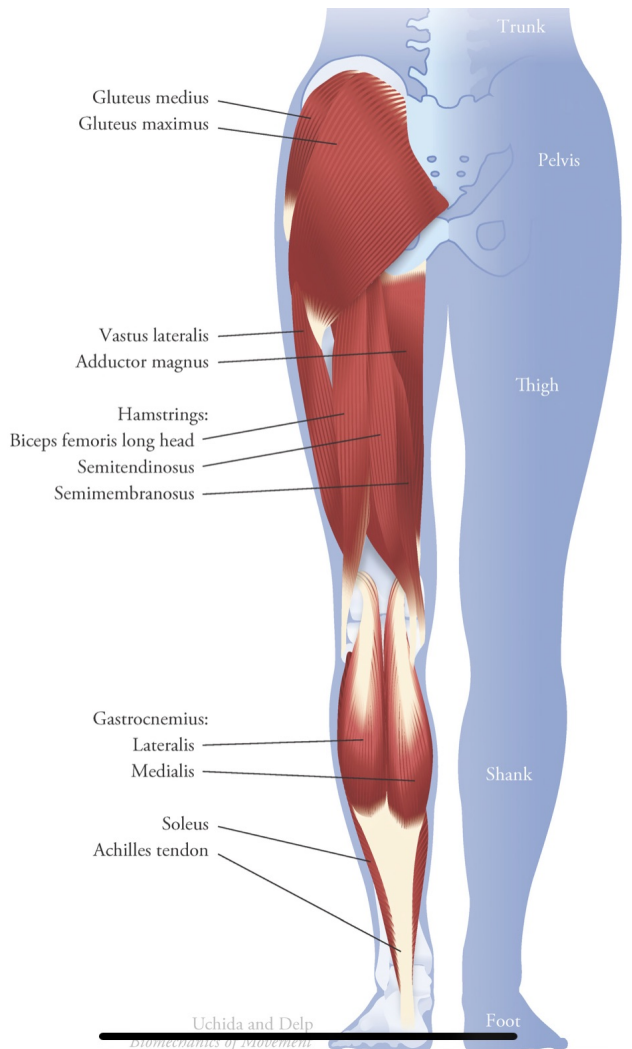


Sagittal plane



Transverse plane





Joints v.s. body segments.

Some body segments:

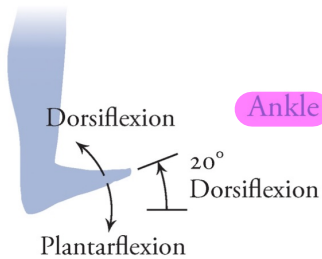
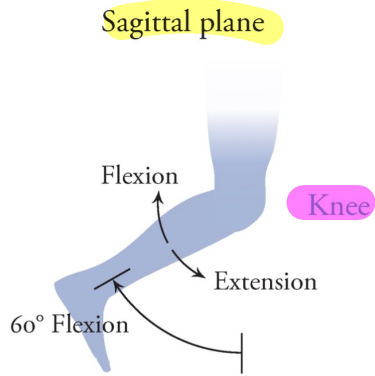
- Trunk
 - Pelvis
 - Thigh
 - Shank
 - Foot
- Angles of body segments are with respect to a global frame, e.g., the gravity vector in the sagittal plane

Joints: (connection point between segments)

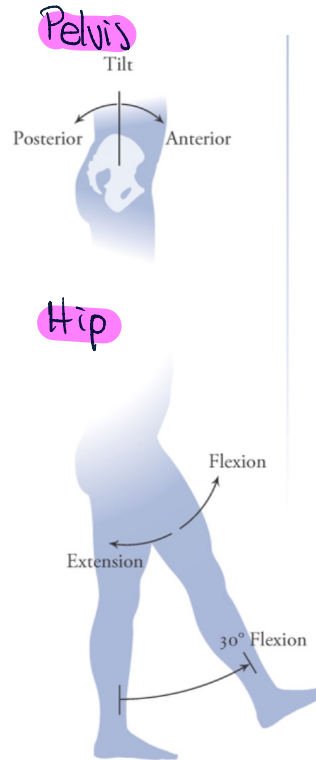
- Ankle
- knee
- Hip

→ Angles are relative to body segments, e.g., Ankle angle is the angle between foot and shank.

Notation: Motion of some joints in the sagittal and frontal planes



Uchida and Delp, *Biomechanics of Movement*



Uchida and Delp, *Biomechanics of Movement*



A note on degrees joint degrees of freedom (DOF)

How many DOFs has the ankle joint?

Some relevant human tissue

- Articular surfaces

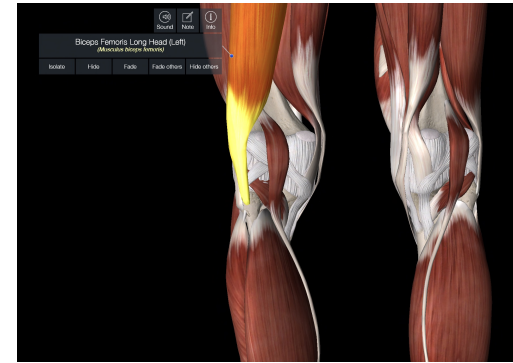
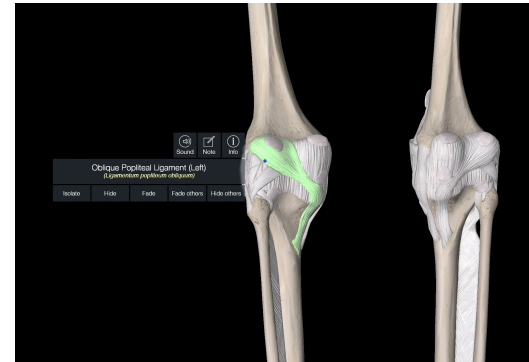
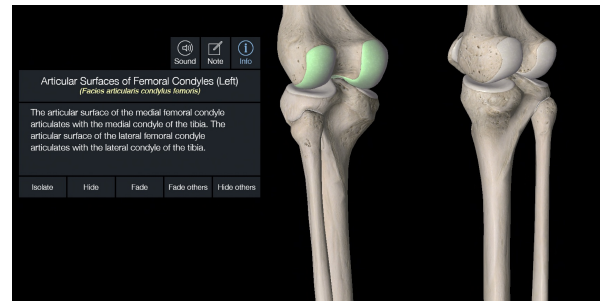
- Ligaments

- Muscle tendon unit

Dr. Gerard Agop Ateshian

08/29/2023 - AME Seminar series

ZPM - Carey Auditorium



Kahoot!

To conclude:

